

CO2 ASSESSMENT TOOL -1

MLA0202-FUNDAMENTALS OF MACHINE LEARNING

Analytical Problem Solving

NAME:K.Hemakshitha

REG NO:192425088

I K.Hemakshitha(192425088) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.Hemakshitha

192425088

QUESTION 1:

CODE:

```
from sklearn.linear_model import LinearRegression
```

```
X=[[1000,2],[1500,3],[800,2],[1200,3],[2000,4]]
```

```
y=[50,75,40,60,90]
```

```
m=LinearRegression()
```

```
m.fit(X,y)
```

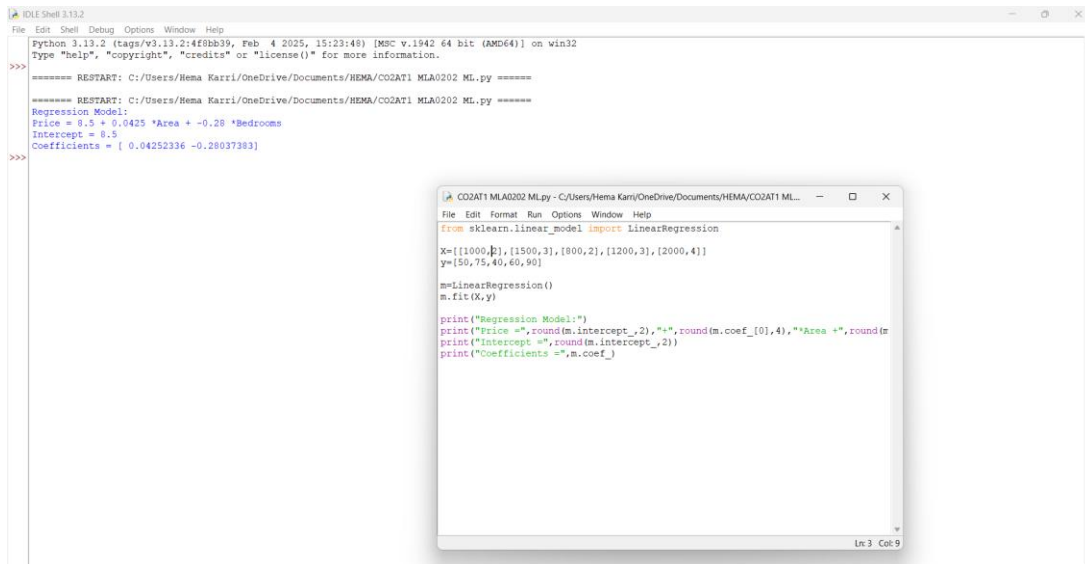
```
print("Regression Model:")
```

```
print("Price =",round(m.intercept_,2),"+",round(m.coef_[0],4),"*Area  
+",round(m.coef_[1],2),"*Bedrooms")
```

```
print("Intercept =",round(m.intercept_,2))
```

```
print("Coefficients =",m.coef_)
```

OUTPUT:



The screenshot shows an IDE window titled 'IDLE Shell 3.13.2'. The main window displays the output of a Python script. The output includes a restart message, a regression model equation, the intercept, and the coefficients. A smaller window titled 'CO2AT1 MLAD202 ML.py' is also visible, showing the source code for the script.

```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb 4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Hema Karri/OneDrive/Documents/HEMA/CO2AT1 MLAD202 ML.py =====
===== RESTART: C:/Users/Hema Karri/OneDrive/Documents/HEMA/CO2AT1 MLAD202 ML.py =====
Regression Model:
Price = 8.5 + 0.0425 *Area + -0.28 *Bedrooms
Intercept = 8.5
Coefficients = [ 0.04252336 -0.28037383]
>>>
```

```
CO2AT1 MLAD202 ML.py - C:/Users/Hema Karri/OneDrive/Documents/HEMA/CO2AT1 ML...
File Edit Format Run Options Window Help
from sklearn.linear_model import LinearRegression
X=[[1000,2],[1500,3],[800,2],[1200,3],[2000,4]]
y=[50,75,40,60,90]
m=LinearRegression()
m.fit(X,y)
print("Regression Model:")
print("Price =",round(m.intercept_,2),"+" ,round(m.coef_[0],4),"*Area +" ,round(m
print("Intercept =",round(m.intercept_,2))
print("Coefficients =",m.coef_)
```

QUESTION 2:

CODE:

```
from sklearn.linear_model import LogisticRegression
```

```
X=[[3,2],[2,1],[1,0],[3,3],[0,1]]
```

```
y=[1,1,0,1,0]
```

```
m=LogisticRegression()
```

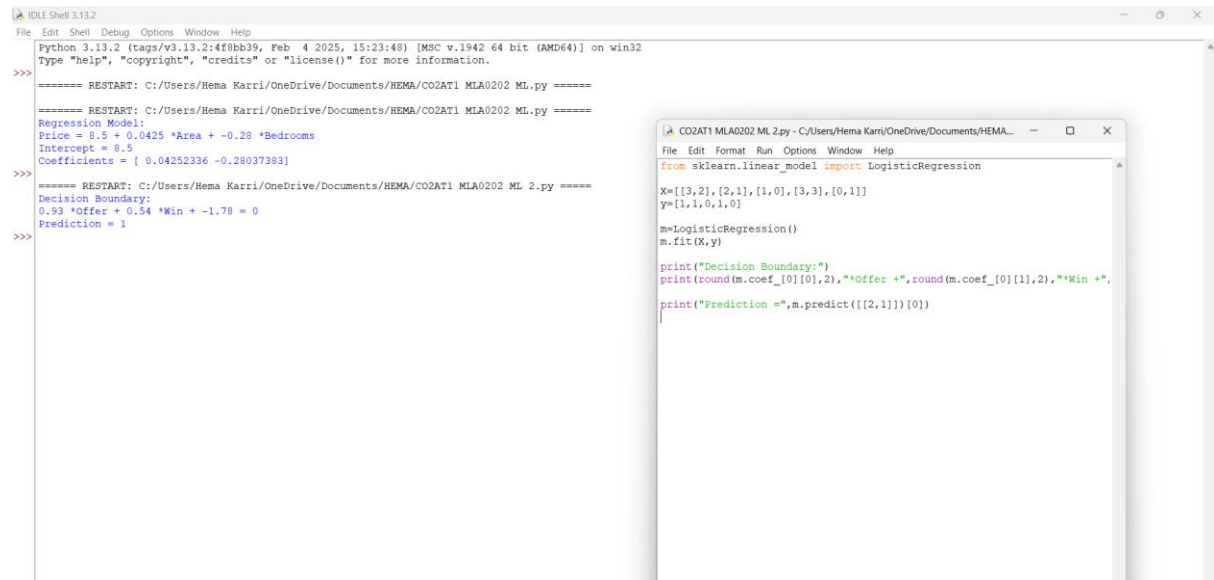
```
m.fit(X,y)
```

```
print("Decision Boundary:")
```

```
print(round(m.coef_[0][0],2),"*Offer +" ,round(m.coef_[0][1],2),"*Win
+" ,round(m.intercept_[0],2),"= 0")
```

```
print("Prediction =",m.predict([[2,1]])[0])
```

OUTPUT:



```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb 4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
===== RESTART: C:/Users/Hema Karri/OneDrive/Documents/HEMA/CO2AT1 ML A0202 ML.py =====
Regression Model:
Price = 8.5 + 0.0425 *Area + -0.28 *Bedrooms
Intercept = 8.5
Coefficients = [ 0.04252336 -0.28037383]
>>>
===== RESTART: C:/Users/Hema Karri/OneDrive/Documents/HEMA/CO2AT1 ML A0202 ML.py =====
Decision Boundary:
0.93 *Offer + 0.54 *Win + -1.78 = 0
Prediction = 1
>>>
```

```
CO2AT1 ML A0202 ML 2.py - C:/Users/Hema Karri/OneDrive/Documents/HEMA...
File Edit Format Run Options Window Help
from sklearn.linear_model import LogisticRegression

X=[[3,2],[2,1],[1,0],[3,3],[0,1]]
y=[1,1,0,1,0]

m=LogisticRegression()
m.fit(X,y)

print("Decision Boundary:")
print(round(m.coef_[0][0],2),**Offer +*,round(m.coef_[0][1],2),**Win +*,
print("Prediction =",m.predict([[2,1]])[0])
```

QUESTION 3:

CODE:

```
py,pn=3/5,2/5
```

```
pfy,phy=3/3,1/3
```

```
pfn,phn=0/2,1/2
```

```
yes=py*pfy*phy
```

```
no=pn*pfn*phn
```

```
print("P(Yes) =",py)
```

```
print("P(No) =",pn)
```

```
print("P(Fever=1|Yes) =",pfy)
```

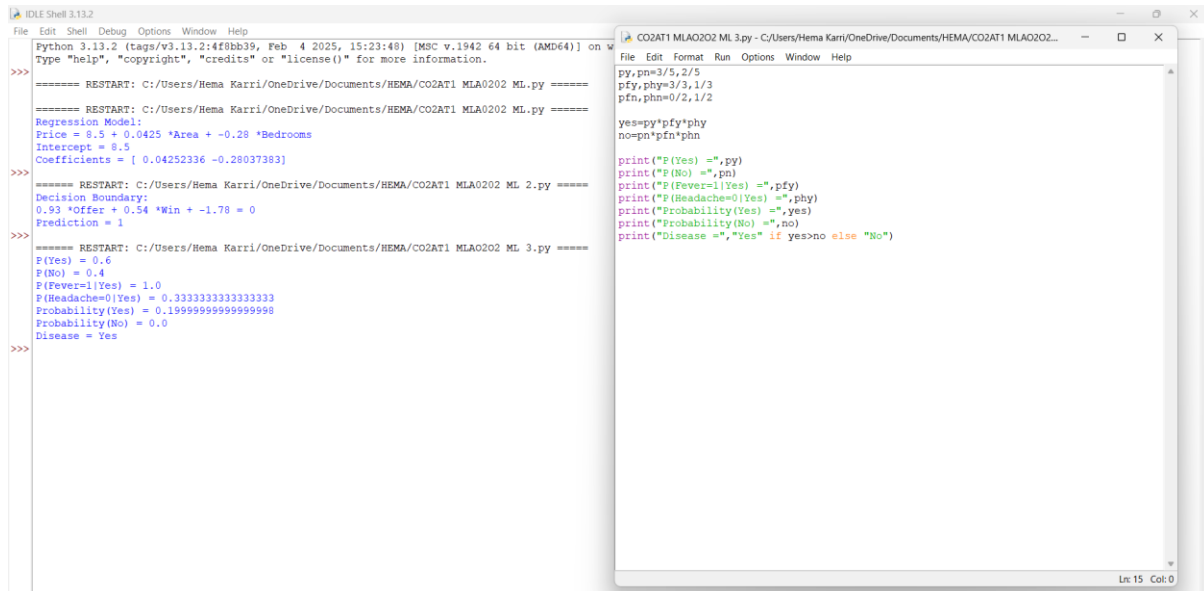
```
print("P(Headache=0|Yes) =",phy)
```

```
print("Probability(Yes) =",yes)
```

```
print("Probability(No) =",no)
```

```
print("Disease =", "Yes" if yes>no else "No")
```

OUTPUT:



The image shows two overlapping Python IDLE windows. The left window, titled 'IDLE Shell 3.13.2', displays the execution of a script. It shows several restarts of the program, with the final output being:

```
>>>
===== RESTART: C:/Users/Hema Karri/OneDrive/Documents/HEMA/COZAT1 MLAO202 ML 3.py =====
Regression Model:
Price = 8.5 + 0.0425 *Area + -0.28 *Bedrooms
Intercept = 8.5
Coefficients = [ 0.04252336 -0.28037383]
>>>
===== RESTART: C:/Users/Hema Karri/OneDrive/Documents/HEMA/COZAT1 MLAO202 ML 2.py =====
Decision Boundary:
0.93 *Offer + 0.54 *Win + -1.78 = 0
Prediction = 1
>>>
===== RESTART: C:/Users/Hema Karri/OneDrive/Documents/HEMA/COZAT1 MLAO202 ML 3.py =====
P(Yes) = 0.6
P(No) = 0.4
P(Fever=1|Yes) = 1.0
P(Headache=0|Yes) = 0.3333333333333333
Probability(Yes) = 0.15999999999999998
Probability(No) = 0.0
Disease = Yes
>>>
```

The right window, titled 'COZAT1 MLAO202 ML 3.py', shows the source code for the script:

```
File Edit Format Run Options Window Help
py,pn=3/5,2/5
pfy,phy=3/3,1/3
pfn,phn=0/2,1/2

yes=py*pfy*phy
no=pn*pfn*phn

print("P (Yes) =",py)
print("P (No) =",pn)
print("P (Fever=1|Yes) =",pfy)
print("P (Headache=0|Yes) =",phy)
print("Probability(Yes) =",yes)
print("Probability(No) =",no)
print("Disease =", "Yes" if yes>no else "No")
```